

Evaluation of Potentially Inappropriate Medications and Polypharmacy in Hypertensive Geriatric Prescriptions Applying Beers Criteria: Consideration of Safer Alternatives in a Multi-Centre Setting.

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1. INTRODUCTION

Geriatrics with hypertension are at increased risk of adverse drug events due to age-related pharmacokinetic changes, comorbidity, and polypharmacy. Potentially inappropriate medications (PIMs) further compromise treatment safety, especially when prescribing deviates from geriatric principles. Age-related pharmacokinetics changes in renal function, slowed hepatic metabolism, and altered body composition heighten the risk of drug accumulation. This study aimed to evaluate the prevalence of PIMs and polypharmacy among hypertensive geriatric patients using the 2023 American Geriatrics Society (AGS) Beers Criteria® and to suggest Safer alternatives using AGS safer alternative list 2025. Pakistan's elderly population is growing rapidly, leading to an increase in age-related diseases such as hypertension. Geriatrics often require multiple medications, making them more vulnerable to polypharmacy and potentially inappropriate medications (PIMs) because of age-related changes in how their bodies process drugs.

2. OBJECTIVES

- To determine the prevalence of potentially inappropriate medications (PIMs) in hypertensive geriatric prescriptions.
- To assess the degree of polypharmacy and its association with PIM occurrence.
- To identify the most commonly implicated drug classes flagged under the Beers Criteria.
- To propose evidence-based, clinically safer therapeutic alternatives for each identified PIM.

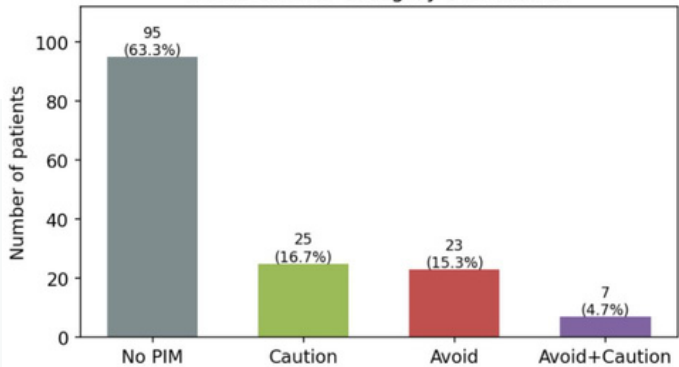
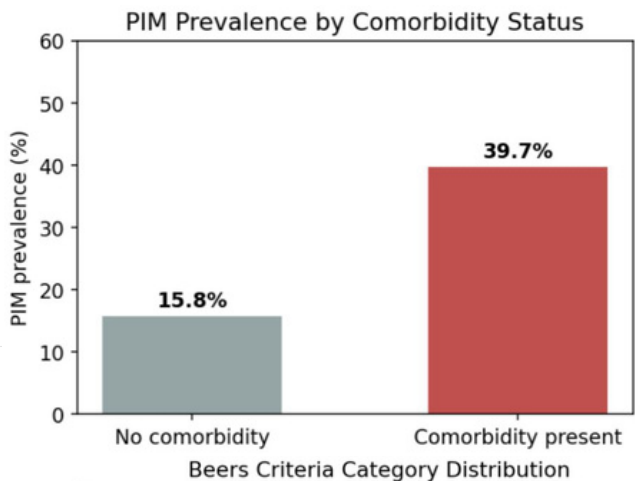
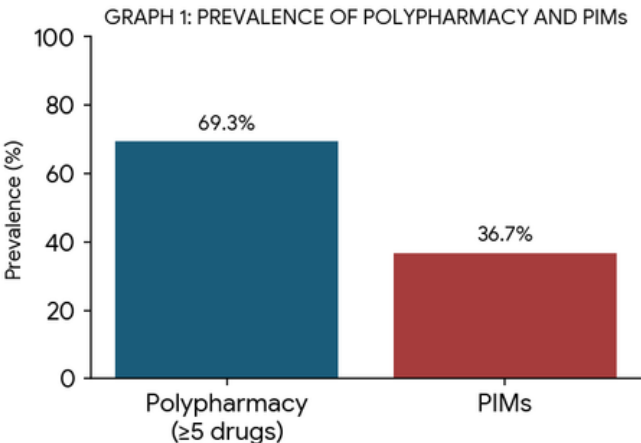
3. METHODOLOGY

This multicenter observational cross-sectional study was conducted across tertiary care hospitals in Lahore, Pakistan, from January to April 2026. Total 150 prescriptions of geriatric patients aged 65 years and above with a confirmed diagnosis of hypertension attending outpatient departments or general medicine wards were collected. Sample size was calculated using the single population proportion formula.

- Population:** Patients aged 65 and older with a confirmed diagnosis of hypertension.
- Criteria:** Screened against the 2023 AGS Beers Criteria across all relevant category (avoid, caution, renal adjustments).
- Polypharmacy:** Defined as concurrent use of ≥5 medications.

4. KEY FINDINGS

- Polypharmacy was highly prevalent, affecting **69.3%** of patients.
- PIMs were identified in **36.7%** of the patients, demonstrating a high burden of inappropriate prescribing.
- A significant association was found between the presence of comorbidities and the occurrence of PIMs ($p = 0.046$).
- Analgesic and central nervous system agents constituted the majority of identified PIMs.



5. DISCUSSION AND CONCLUSION

Both polypharmacy and potentially inappropriate medication use are highly prevalent among hypertensive geriatric patients. The most frequently identified PIMs—NSAIDs (diclofenac, meloxicam), gabapentinoids (pregabalin), benzodiazepines (alprazolam), and SSRIs (escitalopram)—reflect the common treatment of musculoskeletal pain, insomnia, and anxiety in older adults. These medications carry disproportionate risks in patients aged 65 and older, including gastrointestinal bleeding, cognitive impairment, and increased fall risk. The findings strongly support the implementation of systematic, pharmacist-led medication reviews utilizing explicit screening tools like the Beers Criteria to enhance safety.

6. SAFER ALTERNATIVES

Alternatives for the PIMS identified according to AGS Beers Criteria 2023, is suggested form AGS list of safer alternatives 2025 as follows:

Identified PIM	Recommended Alternatives
Diclofenac / Meloxicam (NSAIDs)	Topical NSAIDs, paracetamol, physical therapy; COX-2 inhibitor with PPI if essential.
Pregabalin (Gabapentinoid)	SNRIs (duloxetine), topical agents, CBT. Avoid combined use with opioids.
Alprazolam (Benzodiazepine)	CBT-I for insomnia; SSRIs/SNRIs or buspirone for anxiety management.
Escitalopram (SSRI)	Monitor serum sodium and fall risk; employ non-pharmacological options.

7. Future Directions

Future research should prioritize longitudinal studies to assess the inappropriate prescribing, including adverse drug events and hospitalizations, in elderly hypertensive patients. Larger multicenter studies across Pakistan with complementary screening tools such as Beers Criteria, and integrating clinical parameters, would allow more precise PIM identification. Structured pharmacist-led intervention programs should be developed. Finally, national-level policy initiatives aimed at enhancing clinical pharmacist roles within geriatric care teams could help in safer prescribing practices.